

AURORA BOREALIS COIN

ABRS | OPEN-SOURCE PROOF OF WORK



Technical Whitepaper Version 1.2

Transparency and Risk Edition

Post-launch technical and transparency document

Document date: 24 September 2026

Official public launch: 1 September 2026

Network: Native Layer-1 | Consensus: Proof of Work | Algorithm: KAWPOW

OFFICIAL REFERENCES

auroraborealiscoin.com

github.com/auroraborealiscoin/auroraborealis

Technical disclosure - not an investment prospectus or legal opinion

0. Document purpose and status

This document is the post-launch technical, transparency and risk whitepaper for Aurora Borealis Coin (ABRS). It consolidates the network architecture, consensus parameters, monetary rules, public infrastructure, risk disclosures and project transparency principles in a single reference.

This whitepaper is informational. It is not an investment prospectus, legal opinion, offer of securities, promise of exchange admission, guarantee of liquidity or guarantee of financial return. It has not been approved by a competent authority of the European Union or any other jurisdiction.

Document hierarchy

- Consensus source code and the active blockchain state are authoritative for consensus behavior and on-chain issuance.
- Release manifests and published SHA256 hashes are authoritative for identifying distributed software artifacts.
- This whitepaper explains the system but cannot override consensus rules or historical blockchain data.
- Any legal classification stated here is preliminary and must be reviewed by qualified counsel before a regulated offer or admission to trading.

Version information

Field	Value
Document	Aurora Borealis Coin Technical Whitepaper
Version	1.2 - Transparency and Risk Edition
Date	24 September 2026
Network status	Mainnet live
Official public launch	1 September 2026
Core baseline	Aurora Borealis Core 4.6.3

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2. Executive summary

Aurora Borealis Coin (ABRS) is a native, independent, open-source Layer-1 blockchain secured by Proof of Work. The network uses KAWPOW mining, a 60-second target block interval, Dark Gravity Wave difficulty adjustment and a deterministic consensus-defined emission schedule.

Aurora Borealis Core provides full-node validation, peer-to-peer networking, wallet functionality, graphical and command-line interfaces, RPC functions and block-template support. Project-operated public services include a blockchain explorer, mining pool, faucet, website and network telemetry. These services improve usability but are not required for independent validation.

ABRS is intended to be held and transferred on its native blockchain and used for permissionless participation in the network. Holding ABRS does not grant equity, dividends, interest, voting rights, redemption rights or a contractual claim against project operators.

3. Project principles

- Open-source software and independent auditability.
- Deterministic consensus rules enforced by validating nodes.
- Permissionless Proof-of-Work participation.
- GPU-accessible KAWPOW mining.
- Transparent monetary issuance and explicit premine disclosure.
- Publicly verifiable blockchain state.
- Separation of consensus-critical systems from public web infrastructure.
- Conservative deployment and change-control practices.
- No promise of market value, appreciation, liquidity or financial return.

4. Network identity

Parameter	Value
Project	Aurora Borealis Coin
Ticker	ABRS
Network type	Native independent Layer-1 blockchain
Reference software	Aurora Borealis Core
Consensus	Proof of Work
Mining algorithm	KAWPOW
Difficulty adjustment	Dark Gravity Wave, 180-block window
Target block time	60 seconds
Initial block reward	900 ABRS
Subsidy halving interval	2,100,000 blocks
Mainnet P2P port	17333
Mainnet RPC port	17332
Payment URI scheme	auroraborealis:
Protocol version	70028

5. Core architecture

Aurora Borealis Core is the reference full-node implementation. Its principal responsibilities are:

- validation of blocks and transactions;
- maintenance of local blockchain and UTXO state;
- peer-to-peer network participation;
- enforcement of consensus and monetary rules;
- local RPC functionality and wallet operations;
- production and validation of Proof-of-Work block templates;
- graphical and command-line user interfaces.

Consensus-critical behavior resides in Core and is intentionally separated from the Explorer, Pool, Faucet and other public services.

6. Proof of Work and mining

KAWPOW requires miners to perform computational work before a block can be accepted. A valid block must satisfy the current network target and every other consensus rule. Miners receive the consensus subsidy and eligible transaction fees when their valid block becomes part of the active chain.

Mining is permissionless at protocol level. Participants may connect to the official pool, use an independent pool or operate compatible independent mining infrastructure. The presence of a project-operated pool does not grant that pool authority to alter consensus.

7. Difficulty adjustment

ABRS uses Dark Gravity Wave with a 180-block observation window. The algorithm adapts the Proof-of-Work target to changing network conditions so block production remains near the 60-second target over time. Validating nodes calculate difficulty independently from consensus state.

8. Validation and chain selection

Each full node independently verifies received blocks before acceptance. Validation includes, among other checks:

- Proof-of-Work validity and target compliance;
- block structure, size and serialization rules;
- transaction and script validity;
- subsidy, fee and monetary-rule enforcement;
- chain continuity and network-specific parameters.

Nodes select the valid chain according to accumulated chain work. No website, explorer, pool operator or administrator can override the chain accepted by independently validating nodes.

9. Monetary policy and supply

The issuance schedule is deterministic. The initial mining subsidy is 900 ABRS per block and is halved every 2,100,000 blocks. Because rewards are calculated in atomic units, repeated halvings eventually reach the smallest representable non-zero amount.

Supply component	Amount
Mining emission	3,779,999,999.727 ABRS
Founder premine output	210,000,000 ABRS
Treasury premine output	210,000,000 ABRS
Total premine	420,000,000 ABRS
Total theoretical created supply	4,199,999,999.727 ABRS
Unspendable genesis output	900 ABRS
Maximum spendable supply	4,199,999,099.727 ABRS

The maximum spendable supply excludes the 900 ABRS genesis output. The total premine is approximately 10.000002% of maximum spendable supply. These figures must be verified against consensus source code and active chain state before use in regulated or commercial documentation.

10. Premine and allocation transparency

Aurora Borealis mainnet created two consensus-enforced outputs at block 1:

Historical on-chain designation	Amount	Address
Founder allocation	210,000,000 ABRS	AeaBek4B389g2pkxdRgn3SGznQfwJgj9ns
Treasury allocation	210,000,000 ABRS	AeQqDnuyPc1dZt4HRLapm9Vv3zXuBj6pDx

These are historical consensus facts. Any later transfer, contributor distribution, internal economic designation or stewardship policy does not rewrite what was created at block 1. Later movements should be documented separately with transaction identifiers, destination addresses, purpose and authorization context.

Recommended transparency register

- Current balance and movement history for disclosed project-controlled addresses.
- Transaction ID, date, amount, destination and stated purpose for material movements.
- Clear distinction between Founder, Treasury, contributor and independently controlled holdings.
- Publication of any time-based spending policy and any subsequent amendment.

11. Supply verification

At pre-launch certification height 114, the expected and observed UTXO supply were both 420,102,600.00000000 ABRS. This matched the block-1 premine plus mined rewards at that height. Current supply should always be independently recalculated from active chain state rather than inferred from this historical checkpoint.

12. Public infrastructure

Project-operated infrastructure includes:

- public Core mainnet nodes;
- official website and documentation;
- blockchain explorer;
- KAWPOW mining pool with public Stratum access;
- controlled public faucet;
- live network telemetry and software releases.

These services improve discovery, mining access and network observability. They are not required for independent block validation and should not be treated as substitutes for running a full node. Outages or errors in a public service do not automatically imply a consensus failure.

13. Network and service security

- Consensus RPC interfaces should remain restricted and must not be exposed indiscriminately to the public internet.
- P2P networking is exposed through the designated mainnet port while sensitive administrative interfaces remain access controlled.
- Public web services should use HTTPS, reverse proxies, logging, rate controls and staged operational checks.
- Pool and faucet operations require wallet isolation, credential protection, withdrawal controls, backups and incident procedures.
- No operational safeguard eliminates all technical or human risk.

14. Wallet and key responsibility

Control of ABRS depends on possession of private keys. Loss, disclosure or destruction of keys can cause irreversible loss of access. Users must maintain secure backups, verify addresses and protect wallet passwords, recovery material and RPC credentials.

No legitimate project representative needs a user's private key, seed phrase or wallet backup. Transactions are normally irreversible; mistaken destinations, phishing and compromised devices may result in permanent loss.

15. Open-source lineage

Aurora Borealis Core contains code derived from established open-source blockchain projects, including Ravencoin Core and Bitcoin Core. Applicable upstream copyright and license notices are preserved. ABRS is an independent network with its own consensus parameters, chain history, branding, monetary rules and infrastructure.

Inherited file names, tests, documentation fragments or internal identifiers may remain where technically appropriate. Their presence does not mean ABRS shares the upstream network or economic policy.

16. Decentralization and governance

Independent participants may:

- run full nodes and verify the chain;
- mine independently or operate independent pools;
- build explorers, wallets and compatible tooling;
- inspect source code and reproduce consensus analysis;
- propose, review or reject software changes by choosing which valid software to run.

The project team maintains reference software and official infrastructure, but cannot compel independent nodes to accept invalid rules. Material consensus changes require explicit software changes, public documentation, review and coordinated adoption by network participants.

17. Crypto-asset characteristics and holder rights

Characteristic	Disclosure
Native asset	ABRS exists on its own Layer-1 blockchain and is not represented by a smart-contract token on another chain.
Transferability	ABRS may be transferred between valid addresses subject to consensus rules, transaction fees and network availability.
Value	No fixed value, backing, peg or stabilization mechanism is provided. Market value, if any, is determined by independent market activity.
Holder rights	Holding ABRS does not provide equity, dividends, interest, governance votes, redemption rights or contractual claims against operators.
Utility classification	ABRS is not presented in this document as a utility token granting contractual access to goods or services.
Issuer claim	The network does not provide a centralized promise to redeem ABRS for fiat currency, another asset or a fixed amount of value.

This descriptive positioning is not a binding legal classification. Classification may differ by jurisdiction and should be reviewed before any regulated offering, marketing campaign or admission to trading.

18. Software distribution and assurance

- Software should be downloaded only from official release channels.
- Users should compare SHA256 hashes with the official release manifest before execution.
- Source-code availability permits inspection but does not by itself guarantee compiled-binary integrity.
- Antivirus results may change and can include heuristic false positives; a clean scan is not a security audit.
- A vendor's malware determination concerns the submitted artifact and time of analysis, not every past or future build.
- No comprehensive independent audit of every component is claimed by this document.

Security claims should always identify the exact filename, SHA256 hash, release version, analysis date and scope. Reports concerning a different hash must not be presented as evidence about the current artifact.

19. Sustainability disclosure

ABRS uses Proof of Work and therefore requires computational hardware and electricity. Energy consumption depends on active hashrate, hardware efficiency, miner operating patterns and energy sources. Network hashrate alone is insufficient to calculate exact energy use or greenhouse-gas emissions.

This edition does not publish a verified network-wide energy or emissions figure because a complete, independently validated methodology and miner energy-source dataset are not available. The absence of a quantified figure must not be interpreted as zero environmental impact.

Future reporting objective

- Define a documented estimation methodology and reporting period.
- Record network hashrate sources and representative hardware-efficiency assumptions.
- Separate measured data from estimates and confidence ranges.
- Disclose electricity-source assumptions and calculation limitations.
- Obtain specialist review before using sustainability indicators for regulated admission to trading.

20. Risk factors

20.1 Market and liquidity risk

ABRS may have no active market or only limited liquidity. Prices may be volatile, spreads may be wide and holders may be unable to sell at a desired price or at all. Listing can be delayed, refused or withdrawn by a trading platform.

20.2 Technical and consensus risk

Defects in consensus code, cryptographic dependencies, wallet software or protocol implementations could cause loss, chain disruption, incompatible nodes or emergency upgrades. A majority-hashrate attack, selfish mining, deep reorganization or concentrated mining power could affect confidence and settlement finality.

20.3 Wallet and custody risk

Private-key loss, malware, phishing, compromised backups, incorrect addresses or custodial failure can cause irreversible loss. Third-party exchanges and wallets introduce counterparty and operational risk.

20.4 Infrastructure risk

Websites, seed services, explorers, pools, faucets and public nodes may suffer outages, attacks, data errors, resource exhaustion or provider failure. Multiple public services operated by one project can create practical concentration even where consensus remains decentralized.

20.5 Regulatory, legal and tax risk

Laws and regulatory interpretations can change. ABRS may be classified differently across jurisdictions. Trading, custody, promotion, taxation, anti-money-laundering obligations or access may be restricted. Users are responsible for obtaining appropriate legal and tax advice.

20.6 Project and governance risk

Development resources may be limited. Key contributors may become unavailable. Roadmap items may be delayed, altered or cancelled. Disputes, inadequate disclosure, concentration of project-controlled funds or weak operational controls may harm adoption and reputation.

20.7 Software supply-chain risk

Compilers, dependencies, build systems, hosting platforms and distribution channels can be compromised. Published hashes help identify artifacts but do not prove that an artifact is free from vulnerabilities or malicious behavior.

20.8 Economic and mining risk

Mining revenue depends on block rewards, fees, hashrate, difficulty, hardware performance, electricity costs and any market price. Mining can be unprofitable. Future halvings reduce subsidy, and insufficient miner incentives may affect network security.

20.9 Reputation, misinformation and impersonation risk

False claims, obsolete scan results, fake accounts, counterfeit software, misleading promotions or unsupported accusations may affect users and project reputation. Users should verify official domains, hashes and current evidence.

20.10 Unanticipated risk

Not all risks can be predicted. Combinations of technical, market, legal, operational and human events may produce consequences not described in this document, including complete loss of value or access.

21. Risk mitigation principles

- Publish source code, release notes and artifact hashes.
- Use staged deployments, backups, monitoring and least-privilege access.
- Keep consensus RPC interfaces restricted and isolate operational wallets.
- Encourage independent nodes, pools, explorers and technical review.
- Document material treasury movements and project-controlled addresses.
- Correct obsolete or inaccurate public security claims with hash-specific evidence.
- Maintain incident-response and responsible-disclosure procedures.
- Avoid promises of listing, liquidity, price performance or guaranteed returns.

These measures reduce some risks but do not eliminate them.

22. Roadmap and change control

The project roadmap is maintained separately from this whitepaper. Future work remains subject to technical review, security analysis, participant adoption and available development resources. Roadmap language is directional and must not be interpreted as a binding commitment or guaranteed delivery date.

Change-control principles

- Document the problem, proposed change and consensus impact.
- Separate consensus changes from interface or infrastructure changes.
- Use reviewable commits, tests and release notes.
- Publish upgrade requirements and activation conditions where applicable.
- Preserve version history and avoid silently changing historical disclosures.

23. Regulatory positioning

ABRS is described here as a native transferable crypto-asset that is not designed to maintain a stable value by reference to an official currency, basket of assets or external right. It is not presented as an electronic-money token or asset-referenced token, and this document does not present it as a financial instrument.

These statements are informational and do not constitute a definitive legal classification. Before an offer to the public or admission to trading in the European Union, the responsible person should obtain legal review and determine whether a separate Regulation (EU) 2023/1114 disclosure is required.

Items that must not be invented

- legal entity, registered office, management body or LEI;
- competent authority notification or approval;
- CASP, exchange, admission date, issue price or fundraising target;
- audit completion, environmental measurements or insurance coverage;
- legal rights, guarantees or redemption obligations that do not exist.

24. Official references

Resource	Location
Official website	https://auroraborealiscoin.com/
Source repository	https://github.com/auroraborealiscoin/auroraborealis
Software releases	https://github.com/auroraborealiscoin/auroraborealis/releases
Blockchain explorer	https://explorer.auroraborealiscoin.com/
Mining pool	https://pool.auroraborealiscoin.com/
Faucet	https://faucet.auroraborealiscoin.com/
EU MiCA regulation	https://eur-lex.europa.eu/eli/reg/2023/1114/oj

Third-party websites, exchanges, social accounts and scan services are not official merely because they mention ABRs. Users should verify links from the official website and repository.

25. Version history and final notice

Version	Date	Summary
1.0	Pre-launch	Initial technical whitepaper.
1.1	18 September 2026	Post-launch edition with live services and network disclosures.
1.2	24 September 2026	Expanded transparency, holder-rights, security, sustainability, risk and regulatory-positioning disclosures.

Technical parameters should always be verified against the current consensus source code, active blockchain state and exact release artifact. No statement in this document guarantees safety, legal status, exchange listing, liquidity, price appreciation or financial return.

Aurora Borealis Coin (ABRS)
Open-source Proof-of-Work blockchain